



C & I MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/C & I MAINT/2024-25/CIDCS/367682

Date:

01/08/2024(16:51:50)

Subject:REVISION OF LMI ON SECURITY POLICIES FOR CONTROL SYSTEM(DDCMIS) REV NO-04

LMI on security policies for control systems (DDCMIS) has been revised keeping in view of HMI upgradation in St-2 NTPC SIMHADRI (2*500) and addition of member in information security team.

OGN-OPS-C&I-004_Rev01 and revised LMI is attached for reference.

LMI on security policies of control system (DDCMIS) SMPP-ISO-00-LMI/OGN/OPS/C&I/004 REV-04 is submitted to competent authority for kind approval.

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May pl revise as discussed

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Revised copy attached herewith.

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Note No:5

LMI on Security Policies for Control Systems is submitted for approval

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Please confirm whether format is in order

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Front page of LMI changed as per standard format.LMI name also modified.

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Attached LMI File is as per Simhadri LMI Format

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On:21/08/2024(17:59:01) **Ref:**SMTTP/C & I MAINT/2024-25/CIDCS/367682

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Security Policies for Control Systems (DDCMIS)

LMI/OGN/OPS/C&I/004 Rev. No.:4

August 2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI Name	Security Policies for Control Systems (DDCMIS)
2	LMI No.	LMI/OGN/OPS/C&I/004
3	Reference Document	COS-ISO-00-OGN/OPS/C&I/004 REV-01
4	Superseded Document	LMI/OGN/OPS/C&I/004, Rev No: 3.
5	Prepared by	Ajeet Kr Raushan Sr. Manager(C&I)
6	Checked by	Shalini Singh DGM (C&I)
7	Checked by	Jayamol Rachel V HOD (C&I)
8	Reviewed by	Suresh John David GM(O&M)
9	Approved by	Sanjay Kumar Sinha HOP (Simhadri)

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Security Policies for Control Systems (DDCMIS)

1.0 INTRODUCTION

In order to enforce network security in Plant Control Systems consisting of DDCMIS systems and Control networks, security policies and procedures incorporated here-in are to be followed.

These security policies/procedures envisage formation of an Information Security team. Even though different roles have been identified for the individual members of the information security team, more than one role can be performed by the same person. It may be noted that following policies/procedures are only the operation guidelines and advisory steps to ensure maximum data security.

The intent of this Location Management Instruction is to document the policies to be adopted for securing the network as well as the control systems, based on the reference of Operation Guidance Note COS-ISO-00-OGN/OPS/C&I/004, Rev No:1, May 2020 & CC OS IOM Ref No OS-C&I/TA/05 Ver 1.0 dated 14.07.2021 .

2.0 SUPERSEDED DOCUMENTS

SMPP-ISO-00-LMI/OGN/OPS/C&I/004, Rev No: 3.

3.0 REVISION DETAILS

- Annexures 8.1, 8.7 modified as per St 2 HMI upgradation
- Annexure 8.13 - Scope of audit revised as per third party audit recommendation

4.0 SECURITY POLICIES

The policies apply to Plant Control system networks and to the administrators and users of the systems.

4.1 INFORMATION SECURITY POLICY

4.1.0	Team	An Information Security Team must be established to initiate and control the implementation of information security within CS-LAN. The team shall include Chairman, Information security Manager, Information security coordinator and System /Network /database Administrators.
	Applicable Annexure(s)	Annexure-1: Team Members & Responsibilities

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4.2 Fire Wall Policy

4.2.0	Apply Host-Based Firewalls or Port-Filtering	Apply host-based firewalls or port-filtering tools on end systems, with a default-deny rule that drops all traffic except those services and ports that are explicitly allowed.
4.2.1	Deploy Automated/ Scheduled Operating System/Software Patch Management Tools (If available as a part of system)	Deploy automated/ scheduled software update tools in order to ensure that the OS/Firmware and application software & signatures of IPS/IDS are running the most recent security updates provided by the software vendor. Log of the same to be maintained.

4.3 Information Identification and Classification Policy

4.3.0	Maintain Detailed Asset Inventory information	Ensure that the hardware asset inventory records the network address, hardware address, machine name, data asset owner, and department for each asset and whether the hardware asset has been approved to connect to the network.
4.3.1	Use Passive/Active Asset Discovery Tool (if available)	Utilize a passive/active discovery tool to identify devices connected to the CS network and update the hardware asset inventory.
4.3.2	Address Unauthorized Assets	Ensure that unauthorized assets are either removed from the network, quarantined, or the inventory is updated in a timely manner.
4.3.3	Maintain Inventory of Authorized Software	Maintain an up-to-date list of all authorized software that is required in the enterprise for any business purpose on any business system.
4.3.4	Ensure Software is Supported by Vendor	Ensure that only software applications or operating systems currently supported and receiving vendor updates are added to the organization's authorized software inventory. The software which is not receiving updates from OEM (example windows OS) is to be marked as unsupported.
4.3.5	Utilize Software Inventory Tools (if	Utilize software inventory tools throughout the organization to automate the

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	available)	documentation of all software on business systems.
4.3.6	Address unauthorized software	Ensure that unauthorized software is either removed or the inventory is updated in a timely manner
4.3.7	Address unsupported software	Proper up-grade plan for mitigation of Unsupported software with time line to be ensured. Proper backup and restore plan to be drafted for unsupported software to handle crises.
Applicable Annexure(s)		Annexure-2: Plant Control Systems Complete Configuration Diagram Annexure-3: Asset List - Hardware & Software

4.4 Security Policy Review Policy

4.4.0	Policy Revision	The Information Security Policy document will be reviewed on a two yearly basis, or when there are major changes in the business operations.
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4.5 Information Labelling and Handling Policy

4.5.0	Classification	Ensure that the information either in paper form or stored in removable media like CD-ROM, DVD, will be externally labelled (marked) with the appropriate classification.
4.5.1	Disclosure	When not in use, all classified information will always be protected from unauthorized disclosure. When left in an unattended room, such information will be stored in locked away in appropriate containers
4.5.2	Disposal	Physical media (Hard disks, CDs, DVD) when gone bad, should not be sent for repair to third parties and should be properly disposed by degaussing /physical destruction or shredding

4.6 System Planning and Acceptance Policy

4.6.0	System Planning and Acceptance	Before Acceptance of addition of new systems or up gradation and installation of newer versions, ensure that a complete system test is carried out to check for the following: • Capacity Requirement and Performance
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		• Ease of Use • Effect of the new system on the existing systems and overall security of the CSLAN • Agreed set of security controls are in place • Check to see that installation of the new system will not adversely affect existing systems, particularly at peak processing times. • Training and Change Management in the operation or use of the system • Business continuity arrangements
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4.7 Capacity Management Policy

4.7.0	Capacity Analysis	Continuous/ periodic Capacity Analysis to be carried out for Processing power (processor utilization), RAM, Disk space, Network through put.
4.7.1	Capacity Addition & Clearance	Capacity limits are to be monitored; any Overloading should be alarmed and attended by clearing the load or by adding additional capacity.

4.8 Media Handling Policy

4.8.0	Maintain an Inventory of Sensitive Information	Maintain an inventory of all sensitive information stored, processed, or transmitted by the organization's technology systems, including those located on-site or at a remote service provider.
4.8.1	Remove Sensitive Data or Systems Not Regularly Accessed by Organization	Remove sensitive data or systems not regularly accessed by the organization from the network. These systems shall only be used as stand-alone systems (disconnected from the network) by the business unit needing to occasionally use the system or completely virtualized and powered off until needed.
4.8.2	Manage USB Devices/Removable Media (CD/DVD) drives	Disable USB storage in all maintained assets of CS Network. If the USB storage devices are required, enterprise software should be used that can configure systems

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		to allow the use of specific devices. An inventory of such devices should be maintained.
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4.9 Information Security Awareness Policy

4.9.0	Perform a Skills Gap Analysis	Perform a skills gap analysis to understand the skills and behaviours workforce members are not adhering to, using this information to build a baseline education roadmap.
4.9.1	Deliver Training to Fill the Skills Gap	Deliver training to address the skills gap identified to positively impact workforce members' security behavior.
4.9.2	Implement a Security Awareness Program	Create a security awareness program for all workforce members to complete on a regular basis to ensure they understand and exhibit the necessary behaviors and skills to help ensure the security of the organization. The organization's security awareness program should be communicated in a continuous and engaging manner.
4.9.3	Update Awareness Content Frequently	Ensure that the organization's security awareness program is updated frequently (at least annually) to address new technologies, threats, standards, and business requirements.
4.9.4	Train Workforce	Train Workforce on Secure Authentication, Identifying Social Engineering Attacks, Sensitive Data Handling, Causes of Unintentional Data Exposure, Identifying and Reporting Incidents
Applicable Annexure(s)		Annexure-4: Security awareness program (to be Revised yearly) Annexure-5: Security awareness Training details for users

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4.10 Third Party Access Policy

4.10.0	Inventory of Network Boundaries	Maintain an up-to-date inventory of all of the OT network boundaries.
4.10.1	Deny Communication Over Unauthorized Ports	Deny communication over unauthorized TCP or UDP ports or application traffic to ensure that only authorized protocols are allowed to cross the network boundary in or out of the network at each of the organization's network boundaries.
4.10.2	Manage All Devices Remotely Logging into Internal Network	Scan all enterprise devices remotely logging into the organization's network prior to accessing the network to ensure that each of the organization's security policies has been enforced in the same manner as local network devices.
4.10.3	Third Party Access Control	For third parties, the access restrictions should be documented in a formal contract. Lack of a contract may lead to future misunderstanding between the two organizations.

4.11 Change Control Policy

4.11.0	IT Infrastructure Changes	All changes to Hardware/Software in IT infrastructure including servers, workstations, network devices and security devices of CS-LAN will be managed and controlled.
4.11.1	Logical, Graphical and DCS/PLC Hardware changes	All changes are to taken with prior approval as per Change Request Format. (Annexure-06)
4.11.2	Procedures for implementing Change Requests	Proper walkthrough procedures (SOP's) are to be made and maintained for logical modifications of load/download logics into DCS controllers, perform Graphical changes, Perform Hardware changes in DCS systems. All SOP's shall be accessible by C&I team at any point of time.
Applicable Annexure(s)		Annexure-6: Change Request Format

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	Annexure-7: List of SOP's for implementing logic changes in DCS Controllers, Hardware changes, Graphics/HMI changes
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4.12 Anti-virus Policy

4.12.0	Centrally Managed Software	Utilize centrally managed anti-malware software to continuously monitor and defend each of the workstations and servers.
4.12.1	Updates	Ensure that the antivirus software updates its scanning engine and signature database on a regular basis. Period of update not more than 6 months is to be defined and record to be maintained
Applicable Annexure(s)		Annexure-8: Antivirus Update Information

4.13 System Access Policy

4.13.0	Maintain Inventory of Administrative Accounts	Use automated tools to inventory all administrative accounts, including domain and local accounts, to ensure that only authorized individuals have elevated privileges.
4.13.1	Change Default Passwords	Before deploying any new asset, change all default passwords to have values consistent with administrative level accounts.
4.13.2	Use Unique Passwords	Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.
4.13.3	Use Multi-Factor Authentication	Use multi-factor authentication and encrypted channels for all administrative account access.
4.13.4	Use Dedicated Workstations For all Administrative Tasks	Ensure administrators use a dedicated machine for all administrative tasks or tasks requiring administrative access.
4.13.5	Disable Any Un-associated Accounts	Disable any account that cannot be associated with a business process or business owner.

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4.13.6	Disable Dormant Accounts	Automatically disable dormant accounts after a set period of inactivity.
4.13.7	Lock Workstation Sessions After Inactivity	Automatically lock workstation sessions after a standard period of inactivity.
4.13.8	Segment the Network Based on Sensitivity	Segment the network based on the label or classification level of the information stored on the servers, locate all sensitive information on separated Virtual Local Area Networks (VLANs). Applicable to Station LAN network
4.13.9	Enable Firewall Filtering Between VLANs	Enable firewall filtering between VLANs to ensure that only authorized systems are able to communicate with other systems necessary to fulfil their specific responsibilities.
4.13.10	Physical Restriction	Biometric access control to be implemented to CER rooms & Server/Programmer Rooms
4.13.11	Passwords record maintenance	All Servers/workstations/Application Software/network devices passwords are to be stored in confidential location. Accessibility for the same to be restricted to current Information security team only. It should not be shared with any other member internal or external including auditors. A system is to be devised to update the records whenever there is any change in passwords.

4.14 Monitoring Policy

4.14.0	Utilize Synchronized Time Sources	Use common synchronized time sources from which all servers and network devices retrieve time information on a regular basis so that timestamps in logs are consistent.
4.14.1	Activate Audit Logging	Ensure that local logging has been enabled on all systems and networking devices.
4.14.2	Enable Detailed Logging	Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful

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		elements.
4.14.3	Ensure Adequate Storage for Logs	Ensure that all systems that store logs have adequate storage space for the logs generated.
4.14.4	Regularly Review Logs	On a regular basis, review logs to identify anomalies or abnormal events.
Applicable Annexure(s)		Annexure-9: System Logs Abnormality Information

4.15 Incident Handling Policy

4.15.0	Document Incident Response Procedures	Ensure that there are written incident response plans that define roles of personnel as well as phases of incident handling/management.
4.15.1	Assign Job Titles and Duties for Incident Response	Assign job titles and duties for handling computer and network incidents to specific individuals, and ensure tracking and documentation throughout the incident through resolution.
4.15.2	Designate Management Personnel to Support Incident Handling	Designate management personnel, as well as backups, who will support the incident handling process by acting in key decision-making roles.
4.15.3	Standards for Reporting Incidents	Devise standards for the time required for system administrators and other workforce members to report anomalous events to the incident handling team, the mechanisms for such reporting, and the kind of information that should be included in the incident notification. (as per approved NTPC IT security policy)
4.15.4	Maintain Contact Information for Reporting Security Incidents	Assemble and maintain information on third-party contact information to be used to report a security incident.
Applicable Annexure(s)		Annexure-10: Incident Handling Information

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4.16 Information Backup and Restoration Policy

4.16.0	Procedures for taking & Restoring Backup	SOP's for taking & restoring backup from various systems like DCS Controllers, Application software, firmware, Configuration backups of various systems and network. Should cover every component of complete configuration. All SOP's shall be accessible by C&I team at any point of time.
4.16.1	Ensure Regular Automated/Manual Backups	Ensure that all system data is automatically/manually backed up on a regular basis. Log to be maintained with the details of backup and storage locations.
4.16.2	Perform Complete System Backups	Ensure that all of the key systems are backed up as a complete system, through processes such as imaging or OEM provided solutions, to enable the quick recovery of an entire system.
4.16.3	Validate Backup Media	Prepare proper procedure for backup validation and test during appropriate opportunity/ long Shutdown
4.16.4	Protect Backups	Ensure that backups are properly protected via physical security or encryption when they are stored, as well as when they are moved across the network. This includes remote backups and cloud services.
4.16.5	Ensure All Backups have at least One Offline Backup Destination	Backup data and media will be given appropriate physical and environmental protection consistent with the standards applied at the main information processing facility of CS-LAN. All backup will be stored in fireproof cabinets.
Applicable Annexure(s)		Annexure-11: Data backup Records Format Annexure-12: List of SOP's of Backup & Restore procedures for various DCS Controllers/ Application software/ Network Configurations.

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4.17 Network Access Policy

4.17.0	Maintain Standard Security Configurations for Network Devices	Maintain documented security configuration standards for all authorized network devices.
4.17.1	Document Traffic Configuration Rules	All configuration rules that allow traffic to flow through network devices should be documented in a configuration management system with a specific business reason for each rule, with a specific name responsible for that business need.
4.17.2	Install the Latest Stable Version of Any Security-Related Updates on All Network Devices	Install the latest stable version of any security-related updates on all network devices.
4.17.3	Use Dedicated Machines for All Network Administrative Tasks	Ensure network engineers use a dedicated machine, like Network Management Servers, for all administrative tasks or tasks requiring elevated access.

4.18 User Access Management Policy

4.18.0	Role Base password	• Administrator privileges will be given to administrator user only. • Engineer may be given power user privileges. • Operator may be given user privileges according to the requirement of Control System application software
4.18.1	Password change frequency	Schedule change of passwords should be maintained. In case of any incident of password breach, passwords to be changed immediately
4.18.2	confidentiality	Users will be responsible for maintaining the confidentiality of the password and account, and will be fully responsible for all activities that occur under their account.

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5.0 AUDIT

For checking compliance to the security policies & procedures in Plant Control Systems, security audit by a certified auditor (CERTIN certified & having experience of conducting control system security audit during FAT of major DCS vendor) needs to be carried out once in two years. This shall include vulnerability assessment of the workstations/servers and penetration testing of the Station LAN through the firewall from a node outside the network. Where the audit is in the scope of the agency of an ongoing contract, the same shall be ensured to be done by the agency as per the periodicity envisaged in the contract. Scope of work for the audit is attached at Annexure-13.

Suitable actions based on the findings of the security audit shall be taken up for resolution and compliance

6.0 REVIEW

This LMI shall be reviewed by the HOP on a two yearly basis, or as and when necessary.

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7.0 REFERENCES

CIS Controls Version 7.1

<https://cybernetsecurity.com/industry-papers/CIS-Controls%20Version-7-cc-FINAL.PDF>

SI no	Policy	CIS Controls	SI	Policy	CIS Controls
1	4.2.0	9.4	27	4.13.4	4.6
2	4.2.1	3.4/5	28	4.13.5	16.8
3	4.3.0	1.4	29	4.13.6	16.9
4	4.3.1	1.1/2	30	4.13.7	16.11
5	4.3.2	1.6	31	4.13.8	14.1
6	4.3.3	2.1	32	4.13.9	14.2
7	4.3.4	2.2	33	4.14.0	6.1
8	4.3.5	2.3	34	4.14.1	6.2
9	4.3.6	2.6	35	4.14.2	6.3
10	4.8.0	13.1	36	4.14.3	6.4
11	4.8.1	13.2	37	4.14.4	6.7
12	4.8.2	13.7	38	4.15.0	19.1
13	4.9.0	17.1	39	4.15.1	19.2
14	4.9.1	17.2	40	4.15.2	19.3
15	4.9.2	17.3	41	4.15.3	19.4
16	4.9.3	17.4	42	4.15.4	19.5
17	4.9.4	17.5/6/7/8/9	43	4.16.1	10.1
18	4.10.0	12.1	44	4.16.2	10.2
19	4.10.1	12.4	45	4.16.3	10.3
20	4.10.2	12.12	46	4.16.4	10.4
21	4.12.0	8.1	47	4.16.5	10.5
22	4.12.1	8.2	48	4.17.0	11.1
23	4.13.0	4.1	49	4.17.1	11.2
24	4.13.1	4.2	50	4.17.2	11.4
25	4.13.2	4.4	51	4.17.3	11.6
26	4.13.3	4.5			

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8.0 ANNEXURES

8.1 Annexure-01: Information Security Team

Responsibility	MEMBERS
Chairman	HOD(C&I)
Information Security Manager (ISM)	DGM (C&I-HMI)
Information Security Coordinator	DGM(C&I-HMI)
System Administrator	Sr.Manager(C&I-HMI)
Network Administrator	Sr.Manager(C&I-HMI/SG/TG)
Database Administrator	Sr.Manager(C&I-HMI/SG/TG)

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Team Responsibilities:

The Chairman: The Chairman has the overall responsibility for Information Systems Security in CS-LAN. He will be assisted by a team that is responsible for the implementation and monitoring of Information Security activities in CS-LAN.

Information Security Manager: His/her responsibility is to manage the overall Information Security Programme in CS-LAN. Be responsible for maintaining the Information Security Policy and procedures for use in CS-LAN. The documents will be kept up-to-date. Ensure that, when exceptions to the Information Security Policy are necessitated, the risk acceptance process is completed and the exceptions are reviewed and re-assessed periodically. Assist in the formulation of the management's response to the audit findings and follow-up to ensure that the security controls and procedures, as required, are implemented within the stipulated time frame. Co-ordinate or assist in the investigation of security threats or other attacks on the information assets. Approve all information security related operations at the administrative level duly complying with the Information Security Policy. Assist in the recovery of information and information assets from such attacks. Identify individuals responsible for the protection of information and information assets at CS-LAN.

Information Security Coordinator: The responsibilities of the Information Security Coordinator is to ensure classification of information assets is as per the defined Classification criteria and specify the appropriate levels of security for each class of information assets. Assist the users in the development of specific procedures that meet the Information Security Policy for specific products /applications within the CSLAN. Review audit reports dealing with the information security issues and ensure that they are placed before the Information Security Team. Understand, support and abide by CS-LAN Information Security Policy and Procedures. Ensure that the employees and the related third parties working in CS-LAN understand, support and abide by the Information Systems Security Policy and Procedures. Create a positive atmosphere that encourages employees and the third parties working in CS-LAN to report information systems security concerns / incidents to the Information Security Coordinator immediately. Escalate any information security related issues / incidents to the Information Security Manager. Ensure that information systems security reviews are undertaken, whenever required.

Administrators: The Administrators will take responsibility for the implementation of all applicable controls at their respective areas and also their maintenance on a day-to-day basis. The System, Database & Network Administrator will be in-

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charge of the security of all the systems, database, network and communications in Plant Control Systems.

8.2 Annexure-02: Complete Control Systems Configuration Diagram

ANNEXURE -2 : STAGE-1 NETWORK DIAGRAM

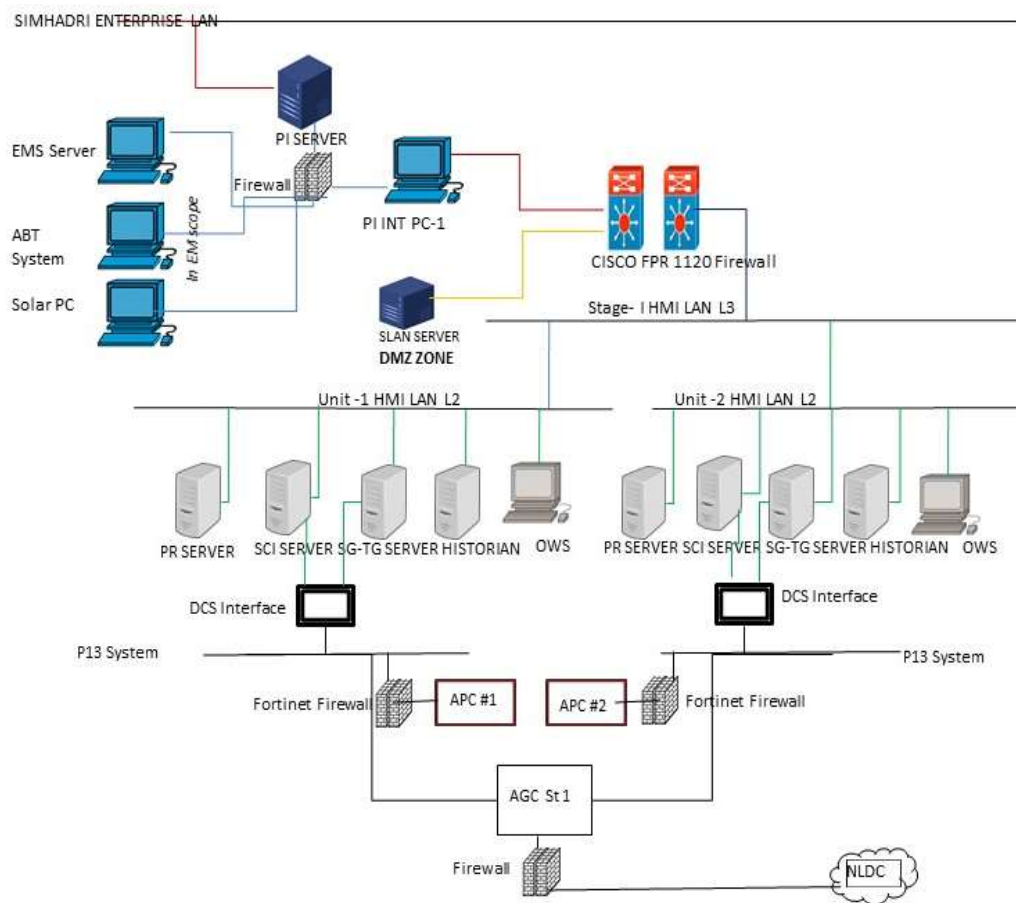


Fig-1

LMI Name	Security Policies for Control Systems (DDCMIS)
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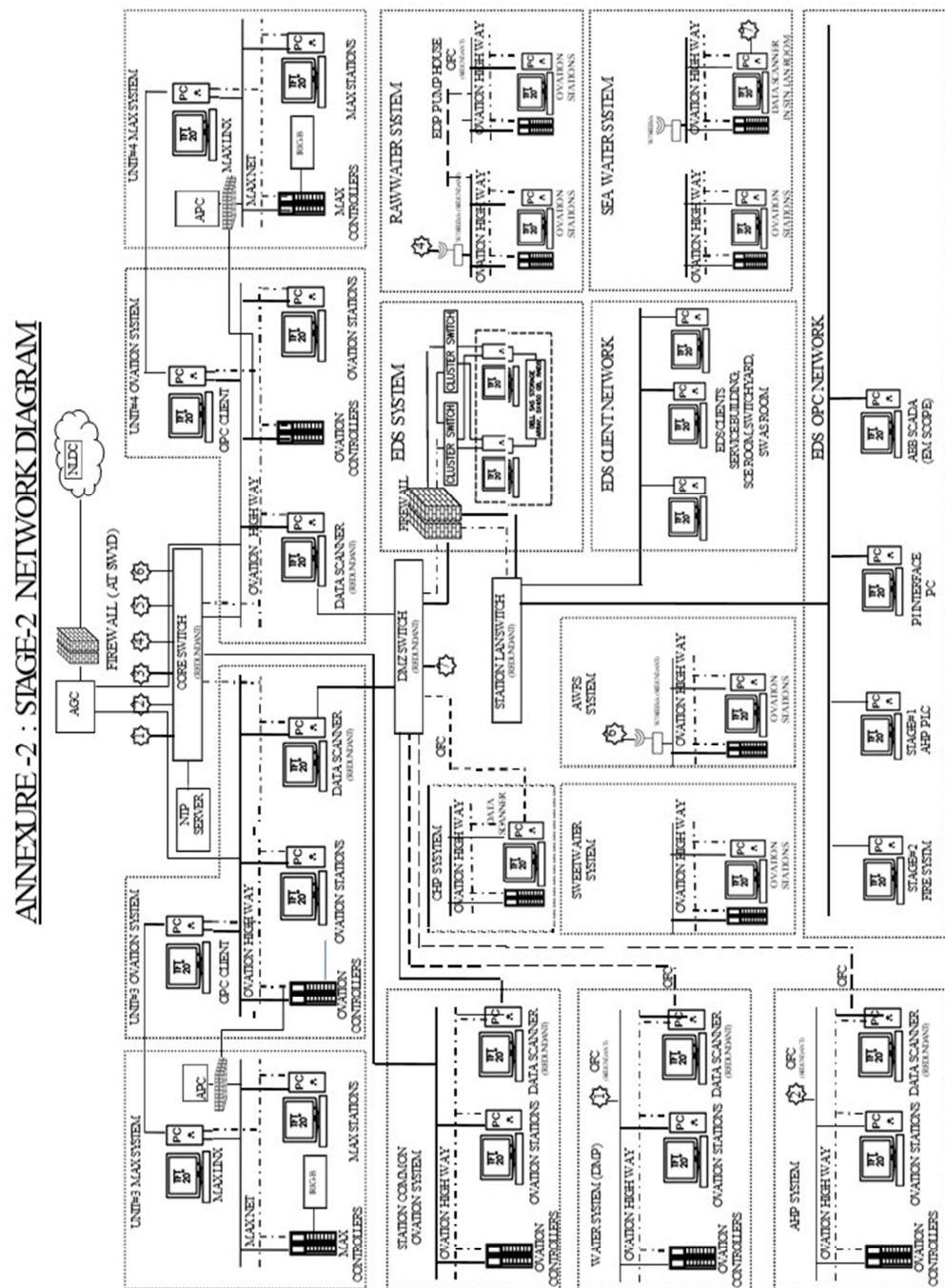


Fig-2

LMI Name	Security Policies for Control Systems (DDCMIS)
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8.3 Annexure-03: Asset List

Authorized Asset: Any asset which is part of the System Configuration diagram/ part of Asset list.

Unauthorized Assets: Assets which are not included in Asset list, upon addition of details will become authorized.

Authorized Software: Software for which License is available or Software which is added Asset list of software

Unauthorized Software: Software for which License is not available or Software which is not part of Asset list.

Unsupported Software: Software for which License is available but not supported by OEM.

Enclosures: Asset Register.

8.4 Annexure-04: Security Awareness Program

Purpose: One principal purpose of information security awareness is to reduce errors and omissions by improving employee compliance and increasing the ability to hold employees accountable for their actions.

Awareness: Stimulates and motivates those being trained to care about security and to remind them of important security practices. Explaining what happens to an organization, its mission, customers, and employees if security fails motivates people to take security seriously.

Program Implementation

Step 1: Identify Training Staff.

Step 2: Identify Target Audiences.

Step 3: Motivate Management and Employees. (Conducting: Quizzes, etc)

Step 4: Administer the Program. (Training, Posters, newsletters, etc)

Step 5: Maintain the Program. (Schedule and implement)

Step 6: Evaluate the Program.

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8.5 Annexure-05: Security Awareness Training Details

S.no	Date	Topic of Training	No of Targeted Audience	No of Audience attended	Feedback Received

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8.6Annexure-06: Change Request (CR) Format

Location/Area:		Date:	
Modification Proposal details: (Attach files if required)			
Details as per LMI: (If applicable only)			
New Requirement:		YES	NO
Existing scheme: (Attach files if required)			
Required Modification: (Attach files if required)			
Attachment List (if any):			
Signatures:(Name, Dept. & Design)			
Change initiator	Approval by Concerned	Chairman	
	HOD	(Information Security Team)	
Implemented By	Information Security Manager		

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Note: In certain cases of modification, some other department may have an affect other than initiating department. In such cases, prior approval of HOD of concerned Dept affected by change to be taken by change initiator before submitting to Chairman.	

8.7 Annexure-07: List of SOPs

For implementing logic changes in DCS Controllers, Hardware changes, Graphics/HMI changes modification in DCS.

Table-1

Sl.no	SOP NO	SOP Description	Version	Update Date
1	DCS/01	01SOP for Procontrol Processor Replacement	1	20-06-2020
2	DCS/02	02SOP for Procontrol DCS logic Modification	1	20-06-2020
3	DCS/04	04SOP for Ovation Logic Modification	1	20-06-2020
4	DCS/06	06SOP for Ovation Controller Replacement	2	15-06-2024
5	DCS/07	07SOP for maxDNA Logic Modification	1	20-06-2020
6	DCS/08	08SOP for maxDNA DPU Replacement	1	20-06-2020
7	DCS/10	10SOP for Graphics Changes	1	20-06-2020
8	DCS/11	11SOP for DCS Simulation	1	20-06-2020
9	DCS/12	12SOP for Network Switch replacement	1	20-06-2020
10	DCS/13	16SOP for S+Graphics Changes	1	25-03-2022
11	DCS/14	17SOP for S+ Database Configuration	1	25-03-2022

Location of Master Copies: C&I WEBSITE->Knowledge Management -> LMI -
><http://191.254.169.3:7878/ciweb/DDCMISSOPS.html>

LMI Name	Security Policies for Control Systems (DDCMIS)
LMI No.	LMI/OGN/OPS/C&I/004 Rev No: 04

8.8 Annexure-08: Antivirus Update Information Format

Sl.no	Date	Definitions update Version	Version Release Date	Status after installation

LMI Name	Security Policies for Control Systems (DDCMIS)
LMI No.	LMI/OGN/OPS/C&I/004 Rev No: 04

8.9 Annexure-09: System Logs abnormality Information

S.no	
Date:	
System Description & ASSET Code:	
Abnormality observed:	
Action Taken:	

LMI Name	Security Policies for Control Systems (DDCMIS)
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8.10 Annexure-10: Incident Handling Information Format

S.n o	Date of Incident	Unit/ Area	Descriptio n of Incident	Threat Level	Logs/Repor ts Storage Location	Correctiv e Measure s Required	Action s Taken	Incident reported to Central cell at

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8.11 Annexure-11 : Data Backup Records Format

S.no	Unit/Area	SYSTEM	Type of Backup	Frequency	Label of Stored device	Date
1.	Stage#1	Procontrol	Logics & configuration	Half yearly		
2.	Stage#1	S+	Graphics & configuration	Half yearly		
3.	Stage#1	S+	OPC Configuration files	Half yearly		
4.	Stage#1	S+	Network Devices Configuration	Yearly		
5.	Stage#2	maxDNA	Historical Pont Configuration	Half yearly		
6.	Stage#2	maxDNA	Logics and configuration	Half yearly		
7.	Stage#2	maxDNA	LVS Alarm, HMI Display Files	Half yearly		
8.	Stage#2	maxDNA	OPC Configuration files	Half yearly		
9.	Stage#2	maxDNA	Network Devices Configuration	Yearly		
10.	Stage#2	Ovation	Historical Pont Configuration	Half yearly		
11.	Stage#2	Ovation	Logics and configuration	Half yearly		
12.	Stage#2	Ovation	OPC Configuration files	Half yearly		
13.	Stage#2	Ovation	Report Server Backup	Half yearly		
14.	Stage#2	Ovation	Network Devices Configuration	Yearly		
15.	Stage#2	Station LAN	EDS Points Back up	Half yearly		
16.	Stage#2	Station LAN	Network Devices Configuration	Yearly		

Note: Data backup media storage are being maintained at two locations mentioned below

1. Electronics Lab - Almirah-1 Rack-1
2. Stage-1 MMI Room Fire proof Rack -1

TAKEN BY:

SIGNATURE

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8.12 Annexure-12-List of SOP's (Backup& Restore)

**List of SOP's of Backup & Restore procedures for various DCS Controllers/
Application software/ Network Configurations**

Table-2

Sl.no	SOP NO	SOP Description	Version	Update Date
1	DCS/03	03SOP for Procontrol DCS Backup&Restore	2	27-12-2021
2	DCS/05	05SOP for Ovation DCS Backup&Restore	1	20-06-2020
3	DCS/09	09SOP for maxDNA DCS Backup&Restore	1	20-06-2020
4	DCS/13	13SOP for Network Switch Backup&Restore	1	20-06-2020
5	DCS/14	14SOP for Stnlan EDS point Backup&Restore	1	20-06-2020

Location of Master Copies: C&I WEBSITE->Knowledge Management -> LMI -
><http://191.254.169.3:7878/ciweb/DDCMISSOPS.html>

LMI Name	Security Policies for Control Systems (DDCMIS)
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8.13 Annexure-13: Scope of WORK for Audit

AUDIT OBJECTIVE:

The major objectives of a security audit are to:

- Identify and understand the existing vulnerabilities
- Provide evidence of compliance with the system security policy.
- Assess the technical and non-technical implementation of the security design.
- Validate proper or improper integration and operation of all security features
- Vulnerability Assessment Testing on Servers.
- Security Audit and Vulnerability Assessment on CONTROL SYSTEM Desktops and Workstations.
- Security Audit on Firewalls, Switches, IDS and wireless Adapters.

AUDIT SCOPE OF WORK:

The Scope involves performing audit in defined procedures & methodologies and reports the vulnerabilities, bugs, and security issues etc., identified during the process based Control System Security policy. Agency shall provide recommendations after auditing based on open and industry . Third party audit can be done only upto HMI but not upto controller level due to system constraints in Stage-1 units.

1. Network architecture design review:

Review the network and system architecture from a security, integrity and availability perspective. The Scope includes providing suggestions on the network design with respect to Control System Security policy against all controls like password management, patch management, ACL etc.,

2. Firewall and Switch Review:

Use automated or manual techniques to check security settings of the routers, firewalls and switches to ensure that they are sufficiently protected from security attacks and hardened. Verification of hardening includes identifying system loopholes in the devices, identifies whether the device is up to date and patched configurations verification etc., with respect to Control System Security policy document and controls.

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3. Internal Vulnerability Assessment and Penetration Testing:

This test shall be conducted on any one system of each network. It shall aim to identify any vulnerability that can be exploited through self or other machines within the local network and not externally reachable. Any software used for testing should be verified on an offline system provided to ensure no effect on Control Network. The scope includes verification of patches, identifying malicious applications and intrusions and recommendations. Check and identify any vulnerability in the operating systems. The checking shall be done in a passive mode in a running unit.

. Active assessment can be carried out only if

a. the unit is in shutdown condition <OR>

b. previous records of successful assessment in unit running condition exist.

In any case it shall be ensured to disconnect the Unit network from the Station network during active assessment.

Penetrating testing is not to be done in the Unit/internal network.

4. External Vulnerability Assessment and Penetration Testing:

Perform network based vulnerability scans to identify security weaknesses of components supporting the critical IT infrastructure and conduct penetration tests in Station LAN Network to identify security holes to the systems connected to external network. Penetration testing can be carried out for the connections from external network.

Eg For PI interface, the penetration testing shall be done from the firewall after OPC Server/

Station LAN Server. Ensure the OPC server/ Station LAN Server is isolated from Unit Network

during the conductance of penetrating testing.

5. Compliance Review:

Evaluates the existing policies and controls and assess the Risk associated.

DEVICES TO BE AUDITED:

- a) Servers
- b) Desktops / Workstations
- c) Firewall
- d) IDS/IPS
- e) Switches
- f) Wireless Adapters

AUDIT PROCEDURE:

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Control System Security audit involves a sequence of steps to be executed by AGENCY in co-ordination with Officials complying the audit objective.

Step 1: Meeting:

Technical Audit Team shall Meet Network and System Administrator team to understand the network architecture and existing Control System Security Policies.

Step 2: Documenting Observations and Understandings:

Technical Audit team shall prepare a document summarizing the points discussed, understood and observations after Step 1. This document will also have a Time-Plan for the entire process and requirements from both the parties.

Step 3: Start the Audit Activities:

The Network and Application Security Audit process starts as per the time plan shared at Step 2.

Step 4: Final Report Submission:

On completing the auditing, a final report shall be prepared by Technical Audit team on the complete list of vulnerabilities and security issues identified during the audit process. Report should include the possible solutions to be adopted to overcome the abnormalities.

Note: Unsupported software can be defined as a “Accepted risk”. Example: Windows XP, Windows 7, Firmware of Network switches, etc., where the up-gradation of complete application software is required for mitigating the risk.

TERMS AND CONDITIONS:

1. A Prior intimation shall be given to the System Administrator about the audit process.
2. The CERT-In certified Auditors only shall conduct the Audit.
3. Meeting shall be conducted in the site. Audit team shall apprise the System administration team regarding the Audit procedure.
4. System administration will allow Technical Audit Team to carry their own laptop and set of tools in CD or USB Drives to the plant premises.
5. Installation of any type of tools / software in Desktop systems and workstations shall be decided after discussion with audit team in order to protect the running plant.
6. The complete information, documents and reports shared among the two parties shall be maintained as confidential documents by AGENCY.

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7. During the Testing Process the network or any other configurations should not be disturbed or updated